Hall Ticket Number:

MR-22

Code: B325CB2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Regular/Supplementary Examinations, December 2025/January 2026

Discrete Mathematics

(Common to CSB and CSD)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Write the following statement in symbolic form. "*If either Jerry takes Calculus or Ken takes Sociology, then Larry will take English.*"	1	2	1	1
1. b)	What is the difference between Statement Calculus and Predicate Calculus?	1	2	1	1
1. c)	Define a set and give two examples.	1	1	2	1
1. d)	Recall the formula for the number of subsets of a set with n elements.	1	1	2	1
1. e)	List any two lattice basic operations.	1	1	4	1
1. f)	Define monoid and state its properties.	1	1	4	1
1. g)	What is meant by repetition in permutation?	1	1	3	1
1. h)	Omit the principle of inclusion and exclusion.	1	1	3	1
1. i)	Tell about Euler circuit.	1	1	5	1
1. j)	Find the purpose of a Hamiltonian circuit.	1	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the basic connectives (AND, OR, NOT, IF–THEN, IF AND ONLY IF) with truth tables.	5	2	1	1
2. b)	Write an equivalent formula for $P \wedge (Q \Leftrightarrow R)$ which contains neither the biconditional nor the conditional.	5	2	1	2
	OR				
2. c)	Obtain the principal disjunctive normal forms of the following: **(a)** $\neg P \vee Q$ **(b)** $(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$	5	4	1	2
2. d)	Let **S(x)** be the predicate "**x is a multiple of 5**" where x is an integer. Express the statement "**There is no integer that is a multiple of 5**" using predicate logic.	5	3	1	2
3. a)	If A={1,2,3,4} and B={3,4,5,6} find A∪B, A∩B, A−B and A∆B using Venn diagrams.	5	2	2	2
3. b)	Define total ordering and partial ordering with one example each.	5	2	2	1
	OR				
3. c)	Let f(x)=3x+2 for x∈R. Show that f is a one-to-one function.	5	4	2	2
3. d)	Given A={1,2,3}, R={(1,1),(2,2),(3,3),(1,2),(2,3)}. Find if R is reflexive, symmetric, or transitive.	5	2	2	2
4. a)	Show with examples how lattice operations satisfy commutative and associative laws.	5	4	4	1
4. b)	Explain the concept of identity and inverse elements in algebraic systems with examples.	5	2	4	1
	OR				

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4. c)	Explain the concept of complement in a Boolean algebra and illustrate with an example.	5	2	4	1
4. d)	Apply Boolean simplification to minimize the expression $F=AB+A^1C+BC$	5	3	4	2
5. a)	Find the number of ways in which the letters of the word 'MATHEMATICS' can be arranged.	5	3	3	2
5. b)	Explain the practical importance of combinatorial analysis in algorithm design.	5	2	3	3
	OR				
5. c)	State and explain the Binomial Theorem with meaning of each term.	5	2	3	1
5. d)	Compute the number of ways to distribute 10 identical balls into 4 boxes such that no box is empty.	5	4	3	2
6. a)	Explain the concept of graph isomorphism with an example.	5	2	5	1
6. b)	Find the chromatic number of a graph with 4 vertices and 4 edges forming a cycle.	5	2	5	2
	OR				
6. c)	Explain the practical significance of the Four-Color Problem.	5	2	5	3
6. d)	Determine if a directed graph can have both Euler and Hamiltonian circuits.	5	5	5	2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome